Sparse Initial Conditions and the Trace Space of Langton's Ant

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Abstract

How many different observations can an ant with at most m initially black cells produce in its first n updates, when the cell coordinates are unrestricted? We prove that the number of anchored turn prefixes is $\Theta_m(n^m)$ for every fixed m. The upper bound encodes a history by the times of its first black reads. The matching lower exponent is realized by a protected single-cell scattering channel, repeated with complete history retained. Every input in this lower-bound family ultimately follows the standard highway. For infinite streams, the countable compact trace spaces obey the exact derived-set identity $\mathcal{T}_m' = \mathcal{T}_{m-1}$: their Cantor–Bendixson rank is $m+1$. An explicit compact subfamily of eventually standard streams has the same rank and prefix exponent. We also determine the optimal passive prediction mistake budget to be m. These results classify sparse-source observation complexity uniformly over unrestricted initial coordinates, transient times and future geometries.

1 The anchored sparse-source question

We use $\mathbb{N} = \{0, 1, 2, \ldots\}$. A classical ant state is a complete finite black set $B \subset \mathbb{Z}^2$, a head position p and a heading $h \in \{N, E, S, W\}$. At each update a white read turns right, a black read turns left, the current cell toggles, and the ant moves one edge. We fix the initial pose to $(0,0), N$; translations and quarter turns normalize all other poses without changing the turn word. Time t is before update t.

Let $\mathcal{T}_m \subset \{\mathrm{R}, \mathrm{L}\}^{\mathbb{N}}$ be the infinite turn streams of complete inputs with at most m black cells. Initial coordinates have no size bound. Let $p_m(n)$ count their length-n prefixes. Let $p_m^H(n)$ count prefixes from those inputs whose future is eventually the standard period-104 highway. A statement about p_m describes the entire sparse-source family, rather than the factor complexity of one selected trace.

Theorem 1.1 (Sharp prefix exponent)**.** *For all nonnegative integers n, m,*

$$p_m^H(n) \le p_m(n) \le \sum_{j=0}^{\min(m,n)} \binom{n}{j}. \tag{1}$$

For every $L \ge 0$ such that $11017 + 208L + 20\min(m, L) \le n$,

$$\sum_{j=0}^{\min(m,L)} \binom{L}{j} \le p_m^H(n). \tag{2}$$

Consequently, for each fixed m,

$$p_m(n) = \Theta_m(n^m), \qquad p_m^H(n) = \Theta_m(n^m), \qquad \lim_{n\to\infty} \frac{\log p_m(n)}{\log n} = m.$$

The exponent is exact; an exact leading coefficient is not claimed. The uniform bounds (1)–(2) also permit m to vary. For $m = 0$ there is just the blank-orbit prefix, so both counts equal one.

Equip the stream space with the prefix topology. For a subspace X, let X' be its set of accumulation points, and define $X^{(0)} = X$, $X^{(j+1)} = (X^{(j)})'$. The rank convention below is the first empty derivative.

Theorem 1.2 (Exact derivative hierarchy)**.** *Each $\mathcal{T}_m$ is countable and compact, and*

$$\mathcal{T}_m' = \mathcal{T}_{m-1} \quad (m \geq 1), \qquad \mathcal{T}_m^{(m)} = \mathcal{T}_0, \qquad \mathcal{T}_m^{(m+1)} = \varnothing.$$

Thus its Cantor–Bendixson rank is exactly $m + 1$.

Theorem 1.3 (Complexity inside the positive basin)**.** *There are explicit compact spaces $\mathcal{P}_m \subset \mathcal{T}_m$, all of whose streams are eventually standard, such that $\mathcal{P}_m' = \mathcal{P}_{m-1}$ for $m \geq 1$, $\mathcal{P}_0 = \mathcal{T}_0$, and their prefixes satisfy* (2)*.*

The last assertion separates observation complexity from convergence: eventually standard inputs already achieve the largest polynomial exponent and derived-set depth allowed by their initial mass. The spaces $\mathcal{T}_m$ are not assumed shift-invariant. In particular we do not identify their anchored prefix exponent with entropy of a bounded-mass dynamical system.

Prior methods and attribution. Gajardo [1] introduced the symbolic read/turn projection for arbitrary colorings and studied its subshift. Same-cell alternation and the corresponding minimal first-read source are established semantics. The compactness and encoding arguments below are elementary persistent-query arguments; the particular dynamics enter the lower bound and the availability of arbitrarily late fresh cells. The 228-update, phase-13 single-obstacle response used here is already present in the one-black project [2]. We reconstruct its complete half-plane form rather than claim a new scattering response or a new one-black theorem. The question addressed here is the sharp all-m census and hierarchy of anchored sparse-source traces, including simultaneous saturation by positive inputs.

2 First-read times encode the entire finite history

For a prescribed word w, its positions and headings are obtained by turning and moving without yet assigning colors. Consecutive reads of the same cell must alternate between R and L. This condition is sufficient: blacken exactly the cells whose first read is L, and start every other cell white. An induction on time verifies the whole word.

Write $M(w)$ for this unique minimal source, and write $E(w)$ for the times of its first black reads. Then $|E(w)| = |M(w)|$. Any other realizing source agrees with $M(w)$ on the complete read footprint, but can contain additional unread black cells. We retain those cells in physical execution.

A finite word is called feasible if it is realized from the fixed initial pose by some finite initial black field. Equivalently, successive visits to each cell request alternating colors.

Lemma 2.1 (Time-code injection)**.** *A feasible length-n word is uniquely determined by $E(w) \subset \{0, \ldots, n-1\}$.*

Proof. Starting from the normalized pose, at an unvisited cell choose L exactly when the present time is in $E(w)$, otherwise choose R. At a visited cell use the opposite of its preceding read. Toggle and move. Each decision and hence every future position is determined. A proposed code marking a revisit is rejected. For a code from an actual word this decoder reproduces the word by induction. □

The upper bound (1) follows by counting subsets of size at most m. This is a time encoding, not a count of coordinate tuples in a radius-n box. It saves one power of n per source cell relative to the crude planar count.

Remark 2.2 (Exactly versus at most). For fixed finite n, the sets of prefixes produced by exactly m and by at most m initially black cells coincide: add padding outside the finite read footprint. This is not an assertion about equality of their infinite trace spaces. A padding cell may be reached later.

3 The complete infinite trace hierarchy

Lemma 3.1 (Bounded-cost branches have one finite source)**.** *Suppose each finite prefix of an infinite word is feasible and has minimal source of size at most m. Then the word belongs to $\mathcal{T}_m$.*

Proof. Minimal sources of nested prefixes are nested: an extension can add a first black cell only outside the earlier read footprint. Their union has at most m elements. On every finite prefix, that union agrees with its minimal source on all read cells. It therefore realizes every prefix and the entire stream. □

Thus $\mathcal{T}_m$ is closed in the compact product space. It is countable because finite subsets of the countable lattice are countable. The infinite union in Lemma 3.1 defines the unique minimal source $M(w)$ of a stream. No algorithm deciding which cells of a given source are never read is assumed.

Lemma 3.2 (Fresh sites occur arbitrarily late)**.** *A classical ant orbit cannot visit only finitely many cells.*

Proof. Otherwise let I be the nonempty finite set visited infinitely often, and choose $v \in I$ maximizing $x + y$. Its north and east neighbors eventually are never visited. The parity of the heading axis plus checkerboard position is invariant; hence every arrival at v has a fixed axis. Late horizontal arrivals must come from the west, heading east. Alternating reads at v eventually force a north departure. Late vertical arrivals must come from the south, heading north, and alternation forces an east departure. Either case contradicts the absent neighbor. This is the usual corner argument. □

This lemma does not claim that individual cells are visited finitely often.

Proof of Theorem 1.2. If $|M(w)| = m$, choose a prefix that has read all its m initially black cells. Any stream in $\mathcal{T}_m$ with this prefix must contain exactly those required source cells and can have no additional first black read. Its minimal source is $M(w)$ and its entire future is fixed by determinism. Hence w is isolated.

If $|M(w)| = r < m$, choose successive fresh cells of w, at times tending to infinity and after all cells of $M(w)$ have been read. Blacken one of them in addition to $M(w)$. The corresponding complete input agrees with w until that first visit, and disagrees at that visit. These distinct streams lie in $\mathcal{T}_m$ and converge to w. Therefore the nonisolated points of $\mathcal{T}_m$ are exactly $\mathcal{T}_{m-1}$. Iteration, with singleton $\mathcal{T}_0$, proves the theorem. □

In particular the point rank of $w \in \mathcal{T}_m$ is $m - |M(w)|$. The prefix tree is decidable by same-cell alternation and first-read counting, and every admitted finite node extends from its finite minimal source. Compactness does not assert that this source is effectively identifiable from an infinite stream after a uniformly bounded observation time.

4 A whole-history protected binary channel

This section supplies the ant-specific lower construction. The finite data are printed in Appendix A and independently replayed. Let

$$t_0 = 11017, \quad p_0 = (-35, -10), \quad d = (-2, -2), \quad q = (-2, -1), \qquad H = \{(x, y) : x + y \le 0\}.$$

Blank execution for t_0 updates has pose (p_0, west) and a complete black field B_* of size 835. Its part in $p_0 + H$ is $p_0 + C$, where C is the displayed 21-cell set. Every other cell of B_* is strictly outside that half-plane. The blank prelude reads none of the ray $p_0 + q + 2id$, $i \ge 0$.

Let w_{start} denote the turn word of the first t_0 blank updates. Let F denote the one-update map on complete states (B, p, h), and write $0 = (0, 0)$ for the coordinate origin. Let W be the

standard 104-letter word in the appendix. The same letter W for a heading is always explicitly called "west" below. Let S be its 228-letter single-cell perturbation word. Exact literal execution verifies

$$F^{104}(C, 0, \text{west}) = ((C + d) \sqcup R, d, \text{west}), \tag{3}$$

$$F^{228}(C \cup \{q\}, 0, \text{west}) = ((C + 2d) \sqcup R_1, 2d, \text{west}). \tag{4}$$

Every read in both equations lies in H. The entire residue R has size 12 and lies in $-4 < x+y \leq 0$. The entire residue R_1 has size 25 and lies in $-8 < x+y \leq 0$. In particular these are complete-field equations, not equality modulo erased debris. Equation (3) implies that W repeats forever, since each new copy stays in a strictly nested half-plane and never reads an earlier residue.

Put $A_0 = WW$ and $A_1 = S$. Both blocks have the same final pose $2d, \text{west}$ and the same retained front $C + 2d$. The zero-block residue is $R_0 = R \sqcup (R + d)$, of size 24; both residues are in $-8 < x + y \leq 0$. Neither block reads any point $q + 2id$ with $i \geq 1$. Both first read q at update 13, where their turns disagree; their preceding turns agree.

Lemma 4.1 (Complete binary composition)**.** *For any finite $J \subset \mathbb{N}$, take the time-zero input*

$$B_J = \{p_0 + q + 2id : i \in J\}, \tag{5}$$

with all other cells white. If $b_i = [i \in J]$, its entire turn stream is

$$w_{\text{start}} A_{b_0} A_{b_1} A_{b_2} \cdots . \tag{6}$$

After L blocks its time and pose are

$$t_L = t_0 + 208L + 20 \sum_{i<L} b_i, \qquad p_L = p_0 + 2Ld, \qquad h_L = \textit{west}.$$

Its complete black field is

$$(B_* \setminus (p_0 + C)) \sqcup \bigsqcup_{i<L} (p_0 + 2id + R_{b_i}) \sqcup (p_L + C) \sqcup \{p_0 + q + 2id : i \in J,\ i \geq L\}. \tag{7}$$

Every such input eventually follows W^ω.

Proof. All input markers avoid the startup footprint. At block i the current protected half-plane is $x + y \leq p_{0,x} + p_{0,y} - 8i$. Every older residue lies strictly outside it. The later markers lie inside but neither local block reads them. Translate (3) twice or (4) according to b_i. The complete residue falls into this block's own eight-level slab; these slabs are disjoint. This proves (7) and the pose/time identities by induction. After the last marker only WW blocks remain. All markers were present at time zero; no intervention or replenishment occurs. □

For $|J| = m$, equation (7) has $835 + 24L + m$ black cells. The original mass is m, whereas the evolving complete mass is unbounded. This illustrates why a sparse-source complexity bound is not a bounded-memory assertion.

Lemma 4.2 (Exact distinction time)**.** *If two binary inputs first differ at slot i, with r common earlier ones, their turn streams first differ at update*

$$t_0 + 208i + 20r + 13.$$

Proof. All preceding blocks coincide. The two current blocks have a common prefix of length 13 and opposite next turns, by the checked kernels. □

5 Sharpness, topology inside the highway basin, and prediction

Proof of Theorem 1.1. The upper bound was proved in Section 2. For the lower bound choose all subsets of $\{0, \dots, L-1\}$ of size at most m in (5). Their first L blocks finish by $t_0 + 208L + 20\min(m, L)$. Lemma 4.2 distinguishes all the resulting prefixes by that time, and Lemma 4.1 proves each input positive. This proves (2). For fixed $m \geq 1$ take $L = \max(0, \lfloor (n - t_0 - 20m)/208 \rfloor)$. The lower term $\binom{L}{m}$ is asymptotic to $n^m/(208^m m!)$ and the upper sum to $n^m/m!$. These prove the sharp exponent, not equality of those constants. □

Proof of Theorem 1.3. Let $\mathcal{S}_m \subset \{0,1\}^{\mathbb{N}}$ contain all binary sequences with at most m ones. It is compact. Equations (5)–(6) define a continuous injection into the turn space: every finite observation depends on finitely many slots. Lemma 4.2 also gives a continuous inverse on the image, which we call $\mathcal{P}_m$. The image is compact and consists entirely of positive inputs. In $\mathcal{S}_m$, a point with exactly m ones is isolated; every point with fewer ones is approached by adding a one at a coordinate tending to infinity. Thus $\mathcal{S}'_m = \mathcal{S}_{m-1}$, giving the same identity for $\mathcal{P}_m$. The all-zero source produces the blank orbit, so $\mathcal{P}_0 = \mathcal{T}_0$. The prefix lower bound is the one just proved. □

Corollary 5.1 (Sharp passive mistake budget)**.** *With unknown initial source and observed past turns, the optimal worst-case number of mistakes of a deterministic next-turn predictor on $\mathcal{T}_m$ is m. This remains a lower bound when inputs are restricted to $\mathcal{P}_m$.*

Proof. The observed turns determine the current position and all earlier reads. Predict the forced color on revisits and R on first visits. Exactly the first reads of initially black cells cause mistakes, at most m of them. For any competing deterministic predictor, choose m successive binary slots. At the decisive update 13 in each slot, choose the source bit opposite to its prediction. The decisions are well-defined from the preceding chosen history. They specify one complete finite seed with at most m black cells, not interventions during execution. Lemma 4.1 realizes it and Lemma 4.2 supplies m mistakes. □

The predictor can store an unbounded visited-cell map. This statement neither supplies a finite-state predictor nor accelerates a known input's simulation. For a supplied n, distinguishing all prefixes needs $m \log_2 n + O_m(1)$ bits of advice: the upper bound encodes first-black times, and the positive lower family gives the matching order.

6 Limitations

The upper bound and derivative identity apply to every complete finite source in the stated mass class, regardless of which HC branch it might follow. They do not assume convergence, classify possible periodic wakes, or require a strip. The protected family is only the witness of sharpness, not a substitute for that all-input scope.

The result is not a termination rank. Shifting the blank stream by four updates gives a word beginning L, whereas the unique member of $\mathcal{T}_0$ begins R. Thus $\mathcal{T}_0$ is not shift-invariant; more generally an update can increase the mass of the complete current state. The derived-set rank belongs to a space of inputs at a specified origin of time. It does not decrease along a physical orbit. Zero exponential anchored prefix rate and finite rank therefore do not imply eventual periodicity.

The theorem does not assert that the full positive subset of $\mathcal{T}_m$ is closed; only the explicitly constructed $\mathcal{P}_m$ is compact. No existence or exclusion of a finite HC counterexample is obtained. The open all-input question still asks whether a single finite source can support infinitely many nonstandard events. The present census answers how many observable histories that family can contain, not whether every one of them standardizes.

7 Reproduction and evidence

Run `python sparse_trace_space/reproduce.py` from the repository root. The default copies only the theorem-local package into a temporary directory, recomputes a deterministic report, and compares it with the frozen one. It does not rewrite evidence. Python standard library and C++17 suffice. The kernels, source fields and complete residues are explicit in the supplement; its source hashes identify the actual checked inputs.

The complete finite first-read tree to depth 24 with source budget 3 has 508 leaves; all encode and decode. Independent heading-index Python and vector-turn C++ simulations compare 1063 inputs, including every one of 511 binary source words of length at most 8, all 512 fields on a 3×3 set and 40 additional sources with remote unread inventory. Each implementation executes 6529739 ordinary updates, and full endpoint coordinate lists agree. All 32640 pairs of eight-bit sources have the exact distinction time. The checks also cover 32 fresh-site approximations, 105 large integer counting identities, three symbolic huge-slot identities and ten malformed inputs. Huge slots are not literally simulated. Infinite and all-m claims are proved in the arguments above, not extrapolated from these controls.

A Complete local data

All listed coordinates are relative to a west-facing head at the origin. The 21-cell front is

$$\begin{aligned} C = \{&(-2,-4),(-2,-3),(-2,0),(-2,1),(-1,-5),(-1,-2),(-1,-1),\\ &(-1,0),(-1,1),(0,-6),(0,-5),(0,-4),(0,0),(1,-6),(1,-5),\\ &(1,-4),(1,-3),(1,-2),(2,-5),(3,-4),(3,-3)\}. \end{aligned}$$

The standard word, with line breaks omitted, is

```
LRLLRRRRLLRLLRRRRLLRRRRLLRLRRRRLRLLLLRRRRLRRLRRRRLLL
LRLRRRRLRRRRLLLLRLRRRRLRLLRRLLLLRRLLRRRRLLRRLRLLRLLR
```

The perturbed word S, with line breaks omitted, is

```
LRLLRRRRLLRLLLLRRRRLRLRRRRLRRRRLLLLRLRRRRLRRRRLLLLRL
RLLLLRRRRLRRLRRRRLLLLRLRRRRLRRRRLLLLRLRRRRLRLLRRLLLL
RRLLRRRRLLRRLRLLRLLRLRLLRRRRLLRLLRRRRLLRRRRLLRLRRRRL
RLLLLRRRRLRRLRRRRLLLLRLRRRRLRRRRLLLLRLRRRRLRLLRRLLLL
RRLLRRRRLLRRLRLLRLLR
```

The complete residue after W is

$$\begin{aligned} R = \{&(-3,0),(-2,-1),(-2,1),(-1,-2),(-1,1),(0,-3),\\ &(0,-2),(0,0),(1,-4),(2,-5),(3,-4),(3,-3)\}. \end{aligned}$$

The complete residue after S is

$$\begin{aligned} R_1 = \{&(-5,-2),(-5,-1),(-5,0),(-4,-3),(-4,1),(-3,-4),(-3,-1),\\ &(-3,2),(-2,-5),(-2,-4),(-2,-2),(-2,2),(-1,-6),(-1,-2),\\ &(-1,1),(0,-7),(0,-3),(0,-2),(0,0),(1,-6),(1,-5),(1,-4),\\ &(2,-5),(3,-4),(3,-3)\}. \end{aligned}$$

These tuples and words determine the full finite checks (3)–(4). The 835-cell startup state is generated by exactly 11017 literal updates from the empty source, rather than specified by a truncated window. Each startup read is tested against the entire marker ray: membership requires equal coordinate differences, divisibility by 4, and a nonnegative index. The two blocks are checked the same way.

References

[1] A. Gajardo. A symbolic projection of Langton's Ant (Extended Abstract). *DMTCS Proceedings AB, DMCS'03*, 57–68, 2003. doi:10.46298/dmtcs.2312.

[2] H. Ke. *Every One-Black-Cell Langton Ant Reaches the Period-104 Highway: A Lean-Checked Proof*. 2026. Article and proof project. The phase-13, 228-step channel is also recorded in the original single-obstacle and separated-obstacle certificate families; the local data required here are fully reproduced in Appendix A.